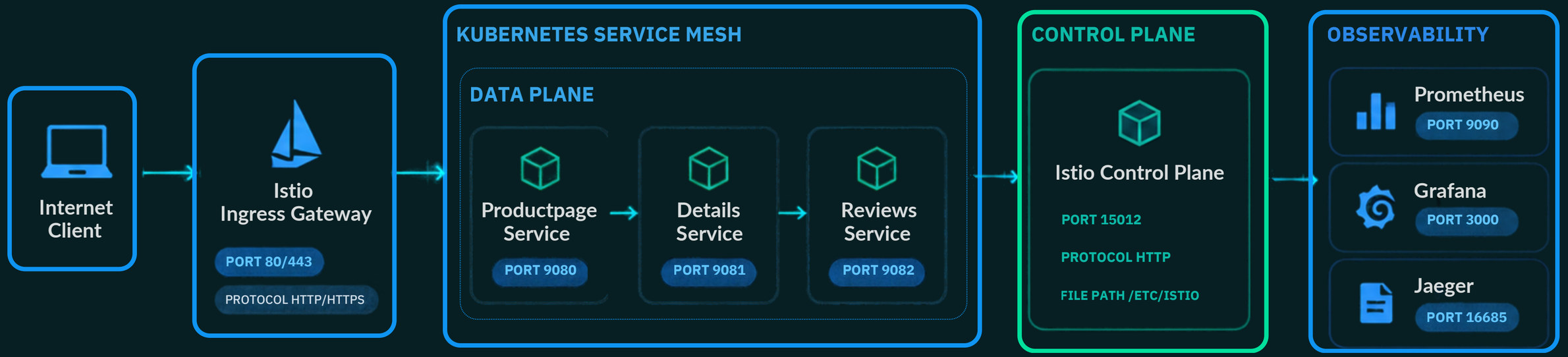
System Architecture —
4-Tier Cloud-Native Architecture Blueprint



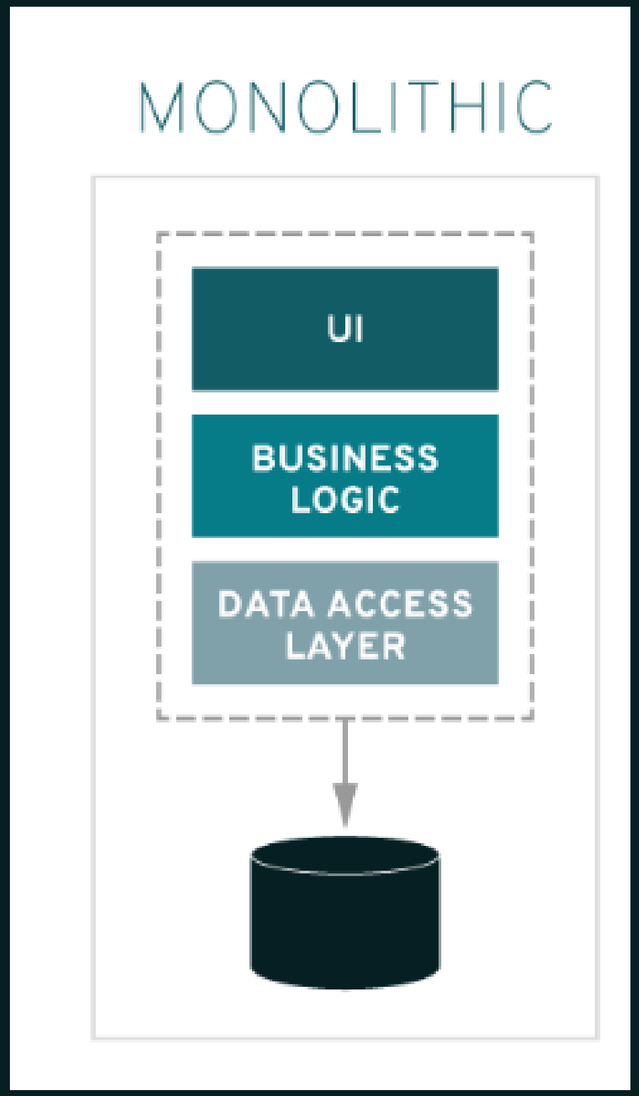
TIER 1
Edge Ingress

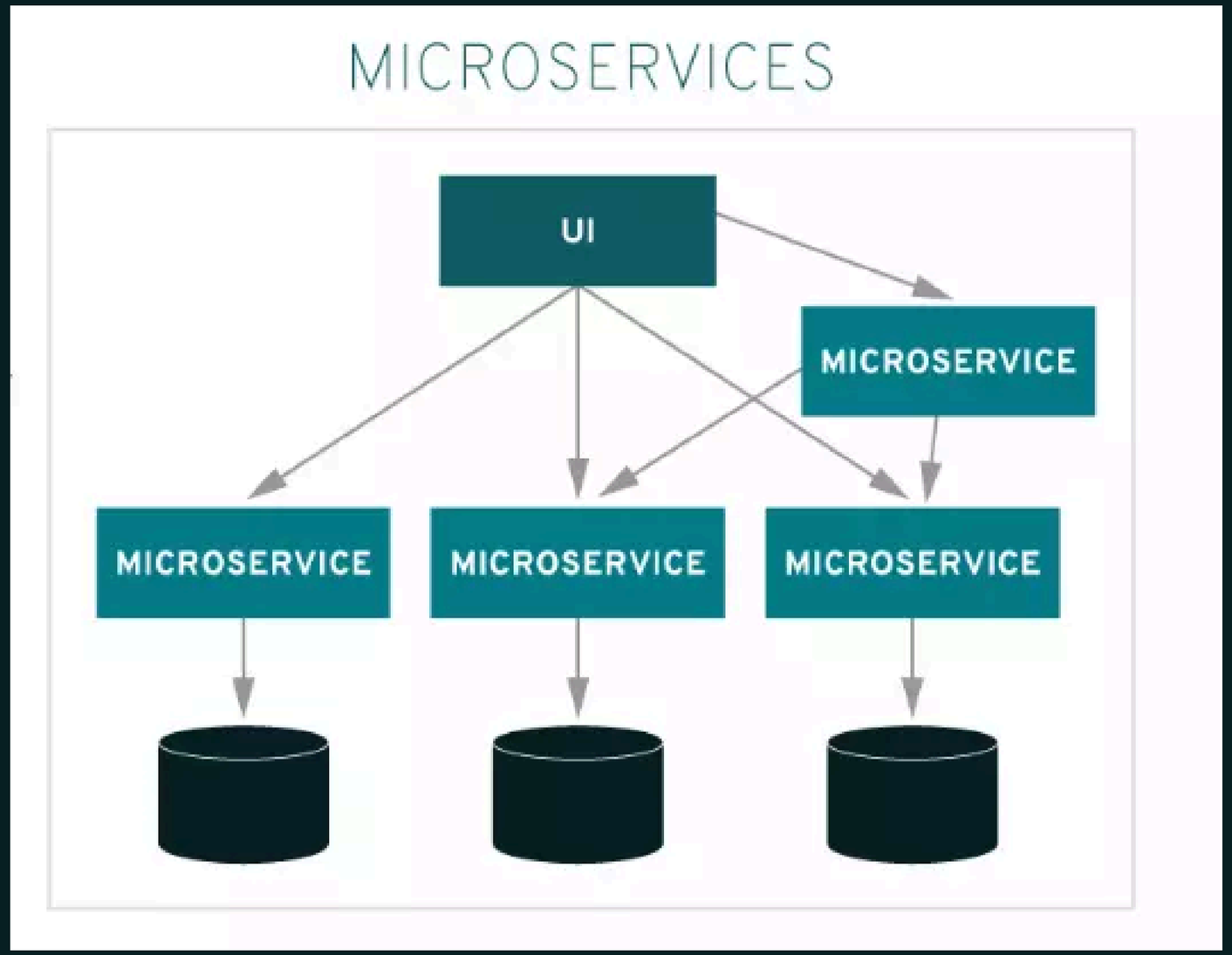
TIER 2
Service Mesh Data Plane

TIER 3
Control Plane

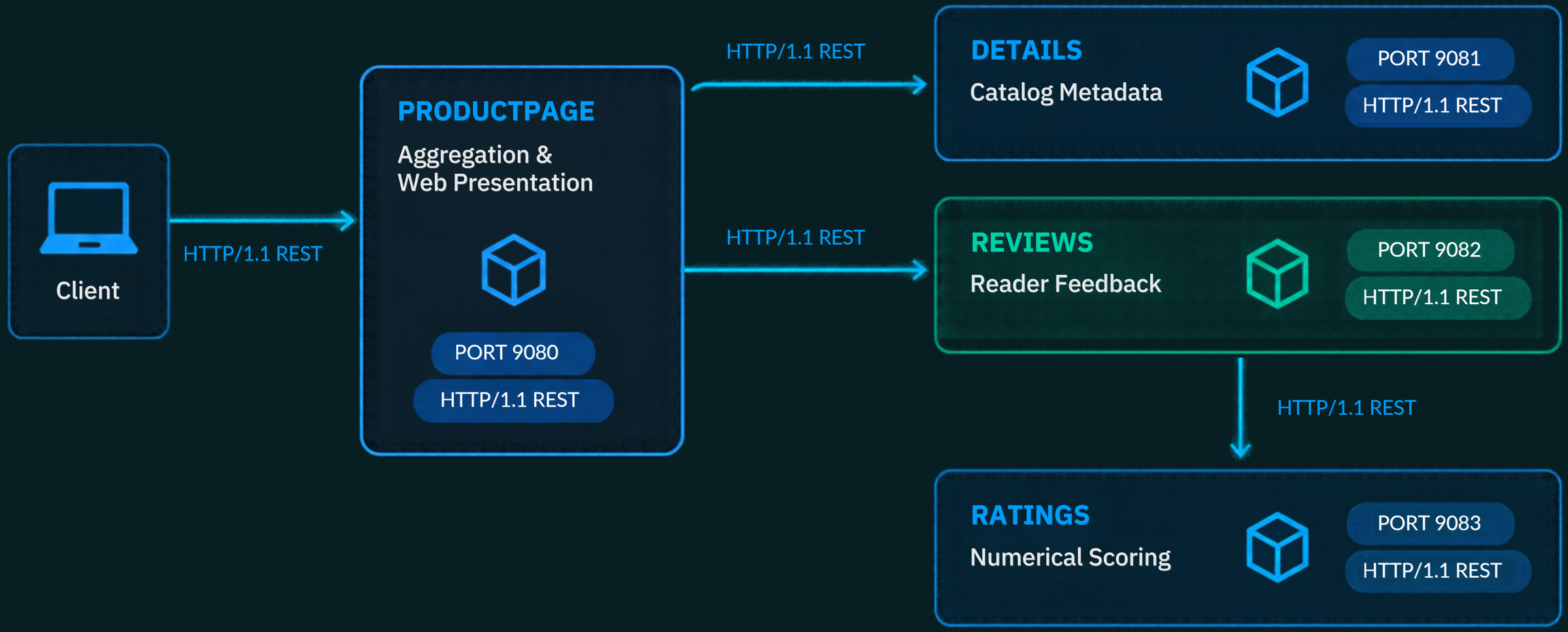
TIER 4
Observability

Microservices Architecture

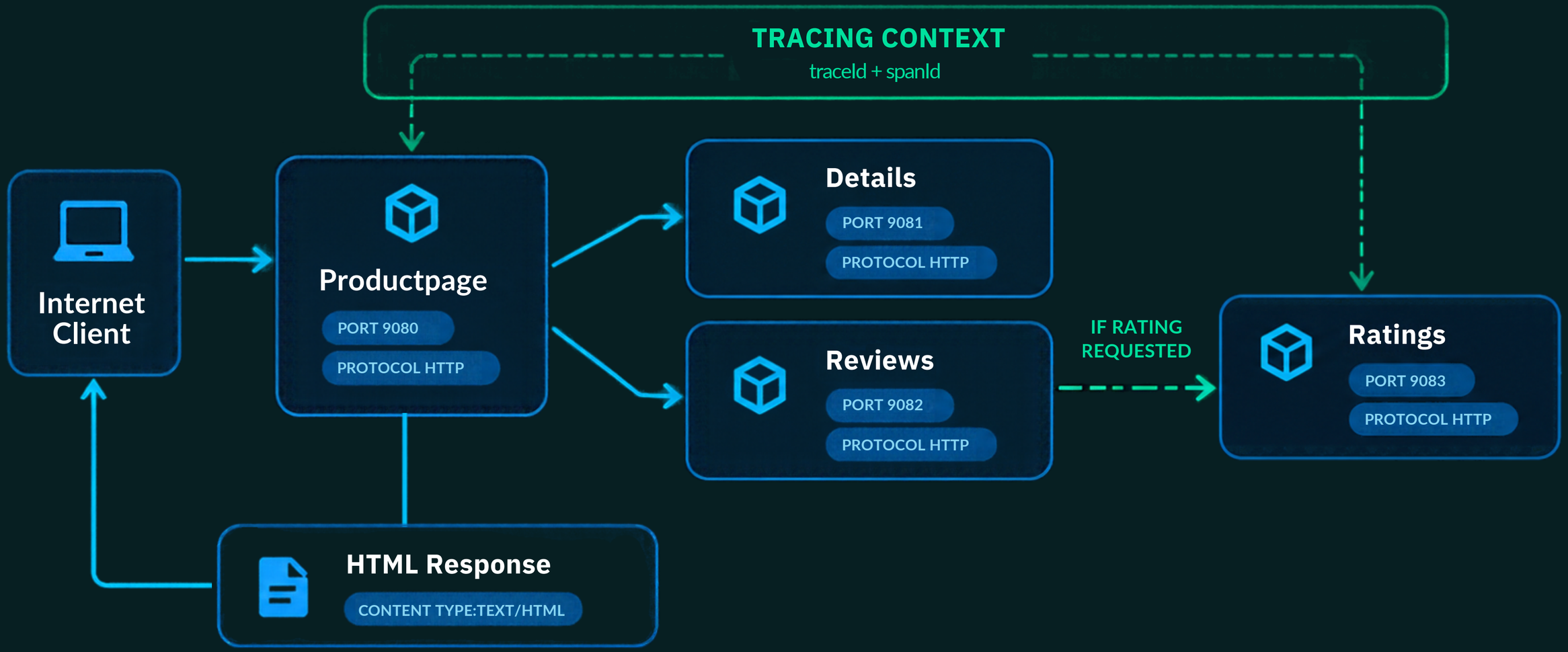




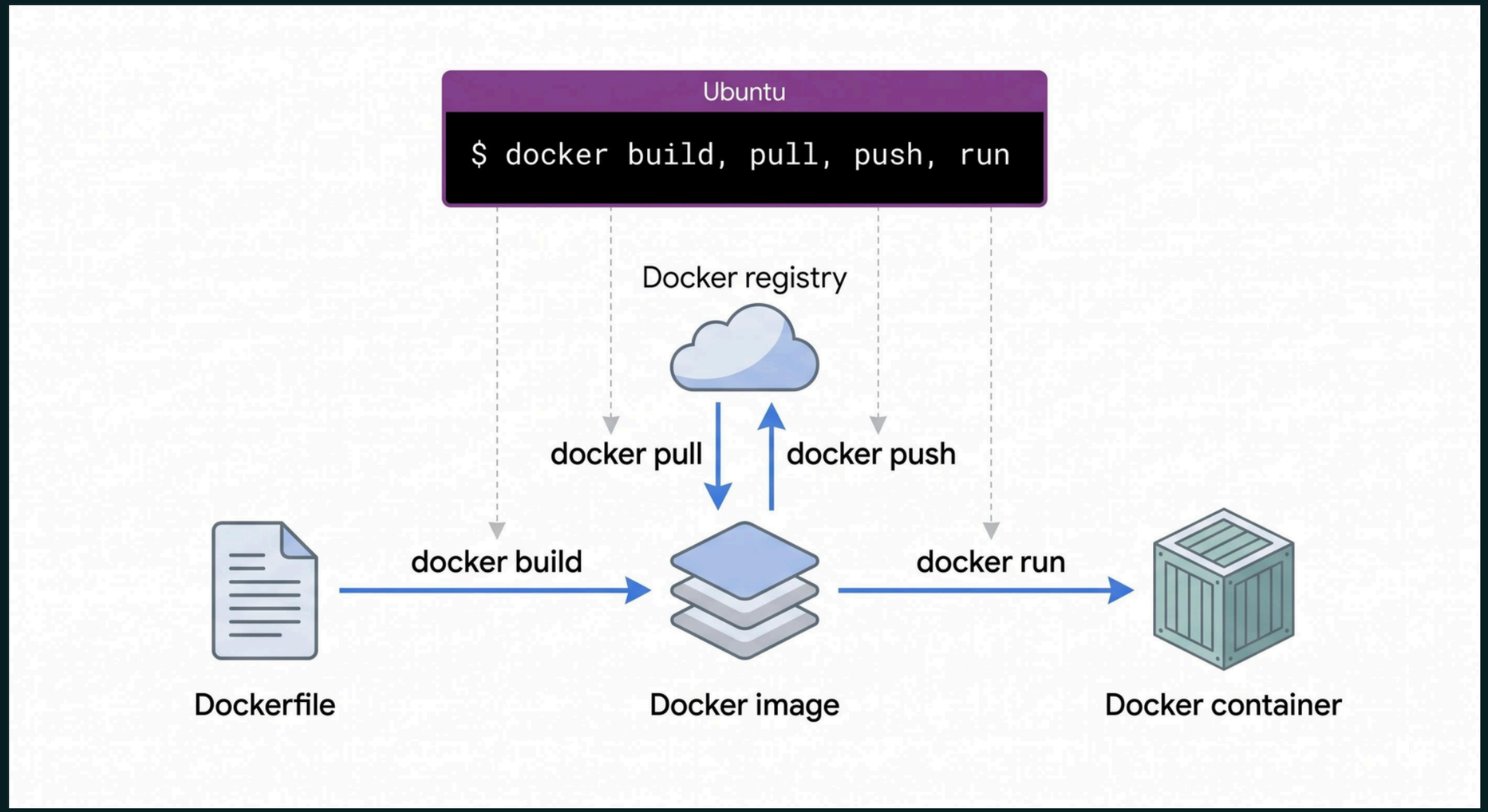
Microservices Architecture —
Domain Decomposition & Bounded Contexts



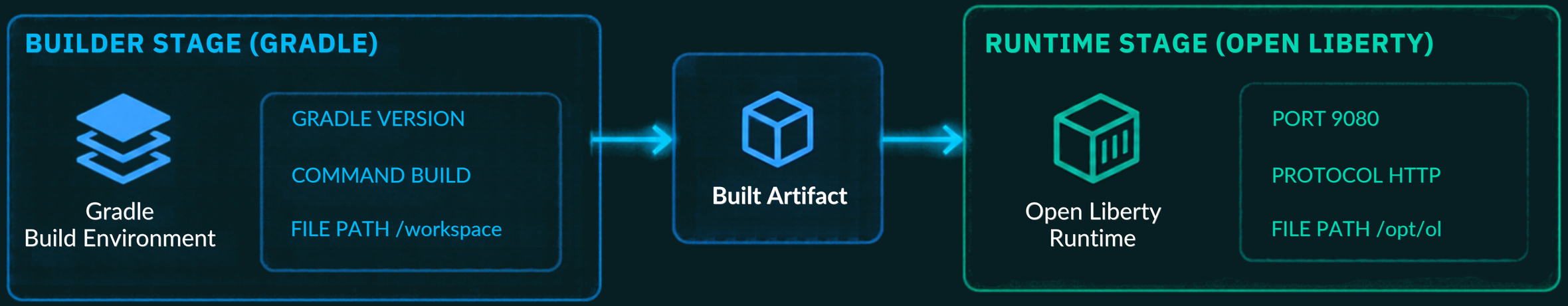
Bookinfo Application —
Polyglot Call Graph & Tracing Architecture



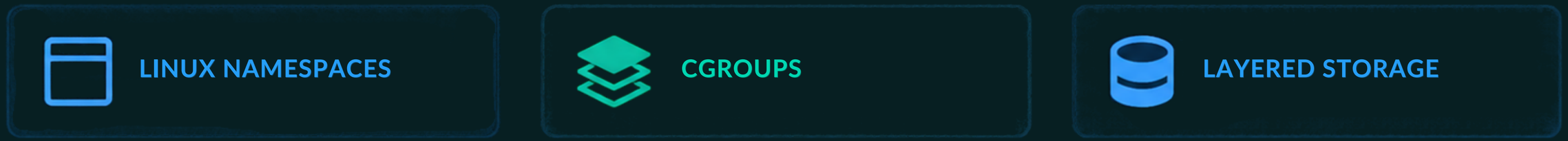
Docker Architecture



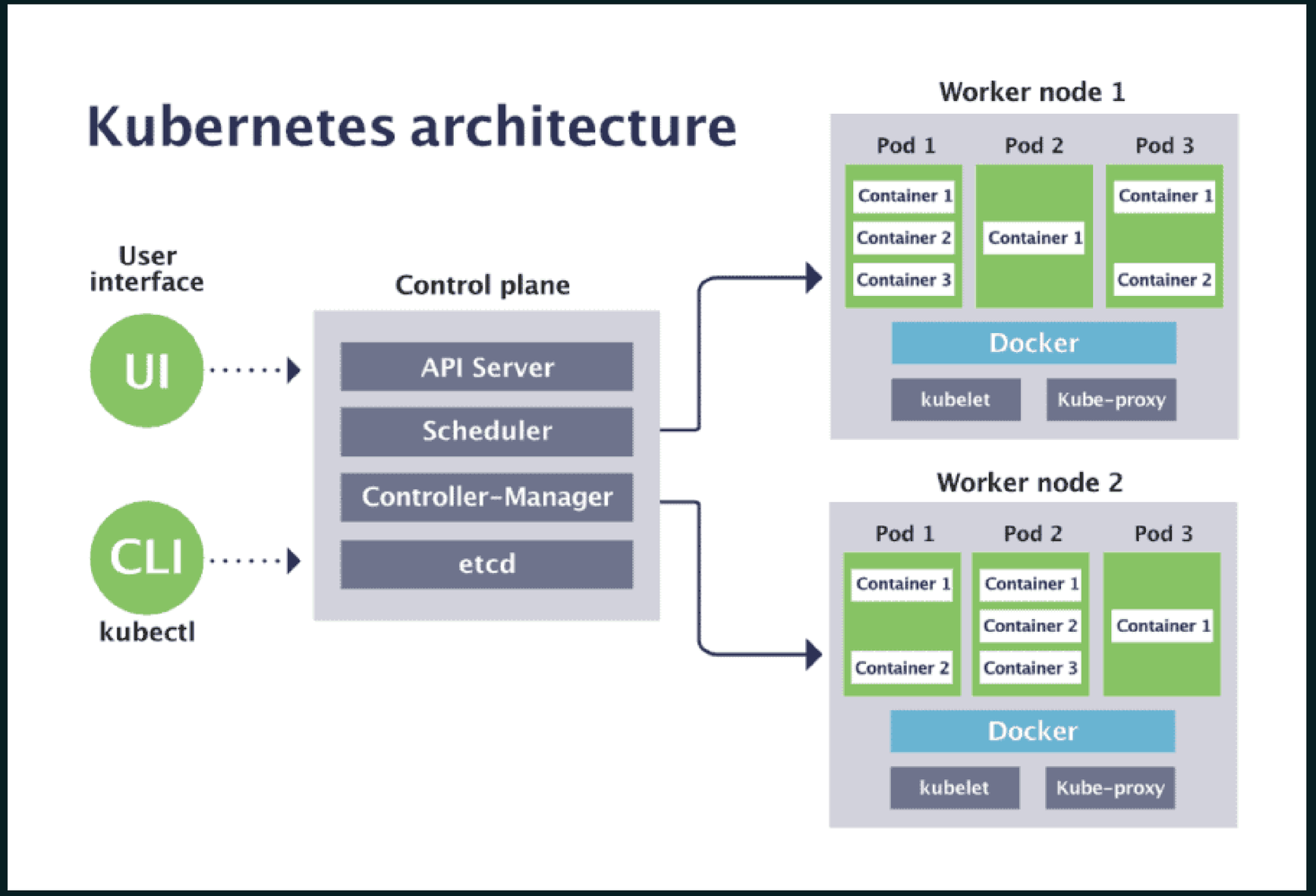
Docker Architecture —
Multi-Stage Build & Kernel Isolation Architecture



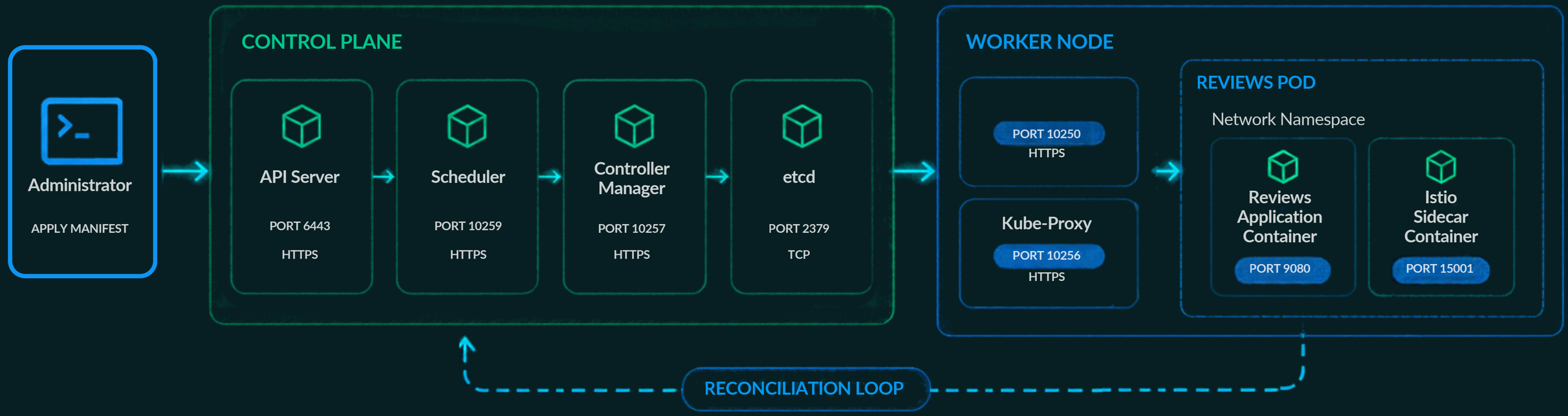
KERNEL ISOLATION



Kubernetes Architecture —



Kubernetes Architecture —
Cluster Orchestration & Pod 2/2 Ready Architecture



1 POD FAILURE DETECTED
Kubelet detects pod failure

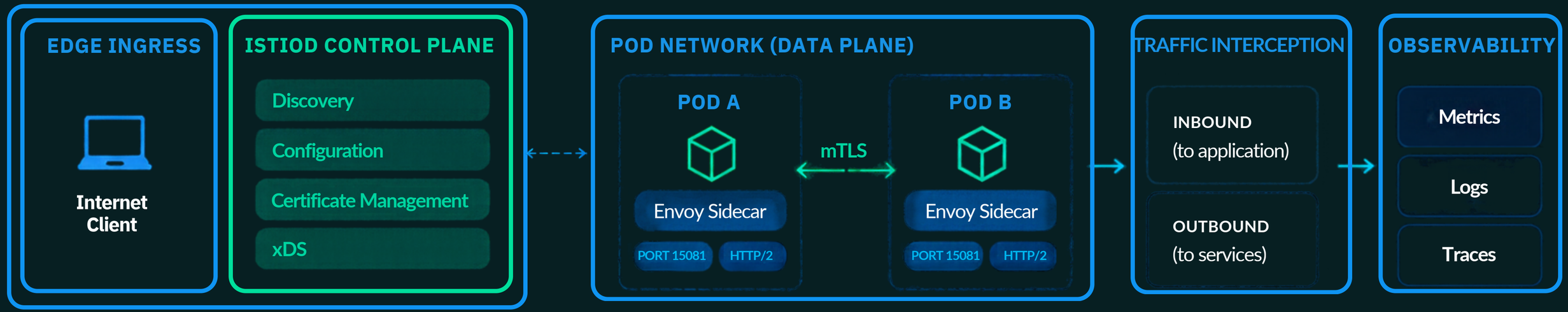
2 RECONCILIATION TRIGGERED
Control plane initiates replacement

3 NEW POD CREATED
Scheduler assigns worker node

4 CONTAINERS STARTED
Application and sidecar initialized

5 READY
Replacement pod reaches readiness

Istio Service Mesh —
Control Plane, Data Plane & Traffic Interception



INBOUND & OUTBOUND TRAFFIC PATHS

INBOUND

Client → Envoy Sidecar → Application

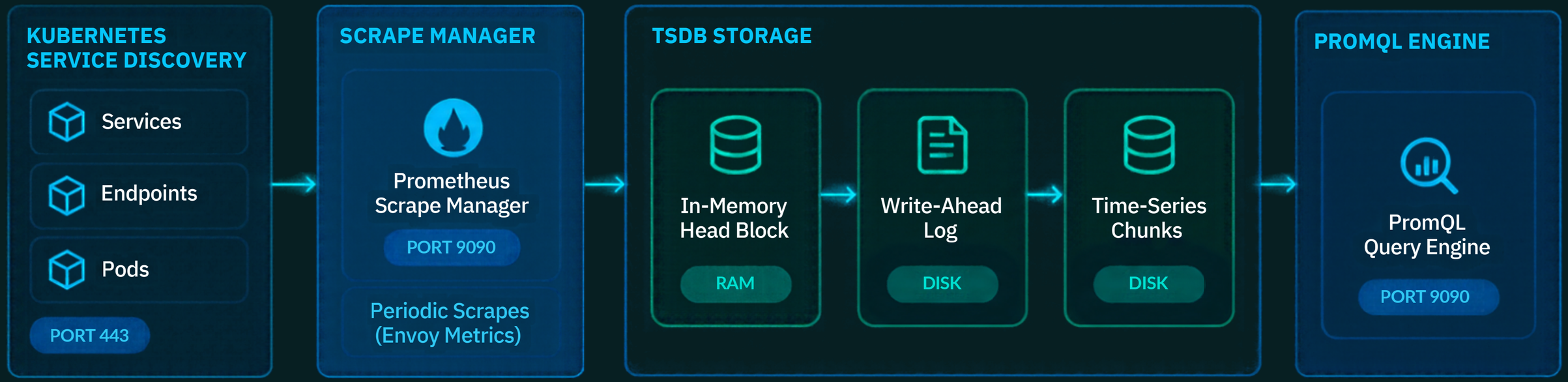
OUTBOUND

Application → Envoy Sidecar → Service

CANARY ROUTING



Prometheus Architecture —
Time-Series Storage (TSDB) & Scrape Engine



Grafana Architecture —
SRE Observability & Telemetry Correlation Architecture

Grafana

Dashboard Filters

PROVISIONED DATA SOURCE

Prometheus

PORT 9990

PORT 9000

PROTOCOL: HTTP

DASHBOARD FILTER PATH

Service → Namespace → Pod → Time Range

GRAFANA DASHBOARDS



SERVICE LATENCY



280ms

SERVICE LATENCY

CONTAINER CPU UTILIZATION



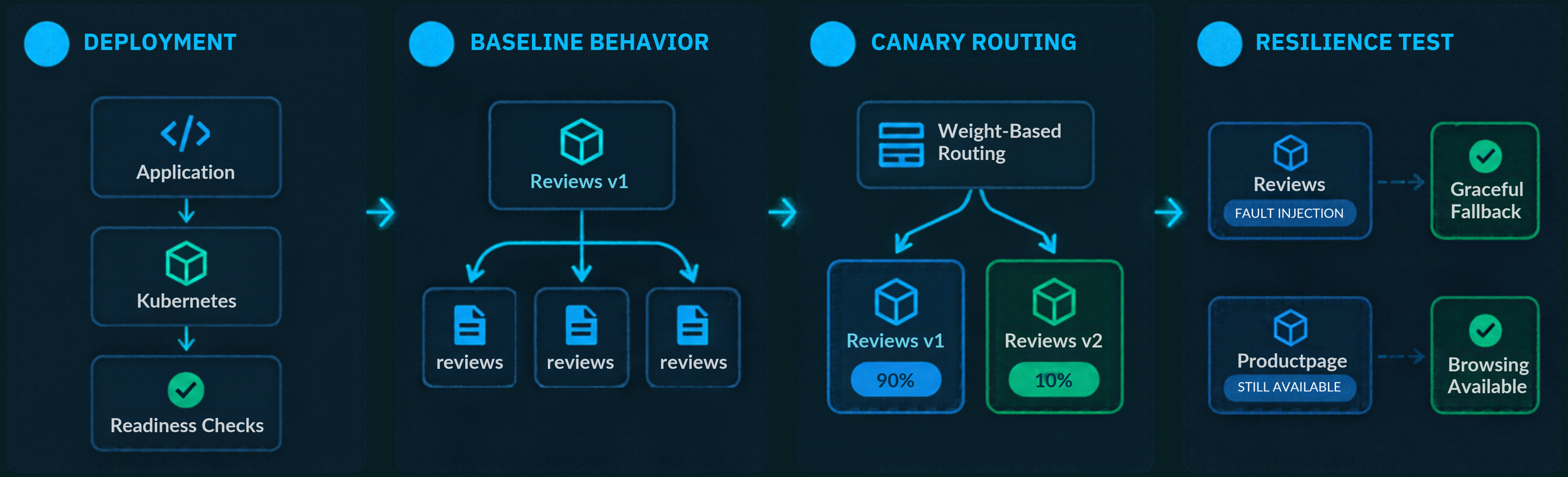
92%

OF LIMIT

DIAGNOSTIC CORRELATION

Increased service latency correlates with container CPU throttling.

Live Verification —
4-Step Architectural Runbook & Chaos Resilience



Architectural Retrospective — Monolith vs Cloud-Native Mesh Trade-offs

	MONOLITH	CLOUD-NATIVE MESH
TECHNOLOGY FLEXIBILITY	Limited	High
DEPLOYABILITY	Slower	Faster
FAULT ISOLATION	Weak	Strong
SECURITY	Higher Risk	Stronger
RESOURCE FOOTPRINT	Larger	Smaller
LATENCY	Higher	Lower
OPERATIONAL COMPLEXITY	Lower	Higher

ARCHITECTURAL VERDICT

Higher operational investment.

BUT THE BENEFITS ARE

- Greater flexibility
- Faster delivery
- Stronger resilience
- Improved security
- More efficient resource use
- Lower latency
- Higher operational maturity